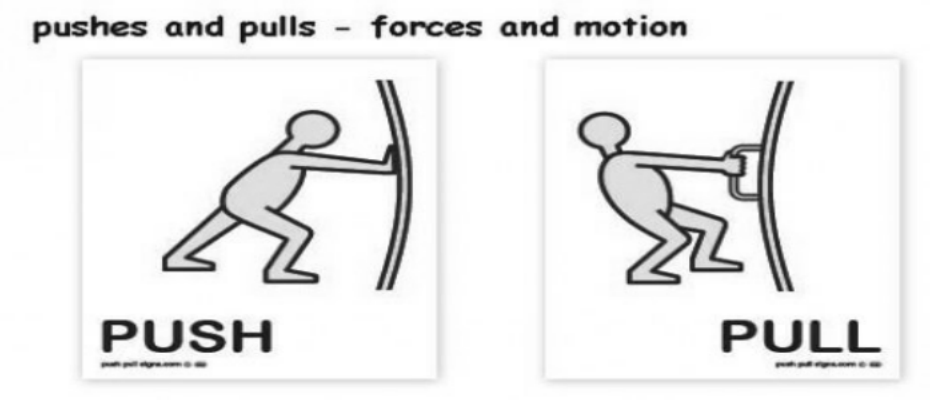
Science Notes

Standard 5 Term 3

FORCE

A force is a pull or push.



Kinds of forces

a) Muscular force

These are forces exerted by people or animals when they use muscles to push or pull.

b) Mechanical Force

These are forces exerted by machines and some kind of motor or engine is used to push or pull.

c) Electric force

Rubbing causes certain things to become electrically charged. These charges exert a force we call an electric force.

d) Magnetic force

The force between the poles of different magnets is called a magnetic force. The North pole and South pole of another magnet attract while the North pole of the one will repel the North pole of the other.

e) Elastic force.

When certain things are stretched or squeezed they pull or push back.

f) Gravitational force

The earth attracts everything and makes it hard to lift things up. This is caused by a force called gravity.

The effects of force

A force can cause two main changes to an object:

- It can change the shape of an object
- It can change the motion of an object

Measuring force

Force can be measured by something that stretches when it is pulled like a spring scale or rrubber band. The larger the force ,the more the spring or the rubber band stretches.

ELECTRICITY

Static electricity is electricity that does not move it stays in one place.

Producing static electricity

- All static electricity is produced by rubbing something.
- Like (same) charges repel one another and unlike (different) charges attract one another.

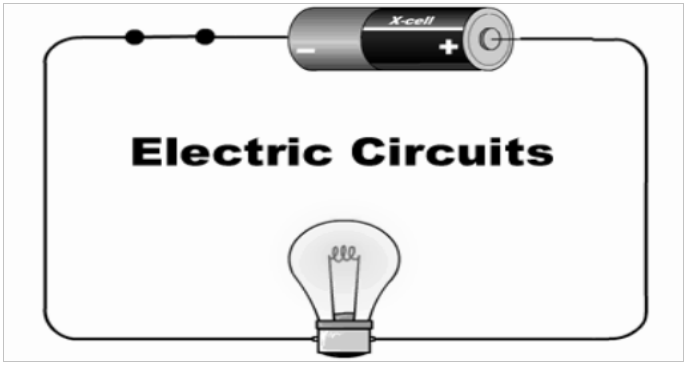
Current electricity

An electric current is a flow of electricity from one place to another along a conductor.

- the electric current is produced by a cell or battery
- electricity flows from positive to negative this is called direct current direct current because t flows to one direction
- the electricity used in homes and shops is different
- it has no positive or negative terminal
- This is called alternating current because (AC) because it changes direction many times each second.

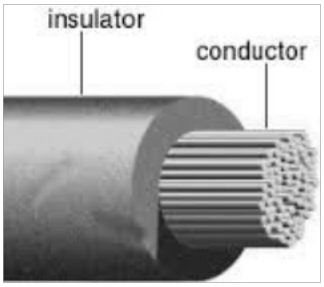
Electric circuits

- an electric circuit must have a source of electricity
- that can be one or more batteries



Conductors and Insulators

- a conductor is material that can carry electricity from one place to the other
- an insulator is a material that will not conduct electricity but will keep in one plac
- Metals are good conductors of electricity.
- Carbon conducts electricity (it is the only metal that does conduct electricity)
- Nonmetals such as wood, plastics, paper, glass, rubber do not conduct electricity.



Current in a circuit

The current in a circuit can be affected by number of cells,

- The number of lamps (globes/light bulbs)
- The strength of an electric current depends on the number of cells in a circuit

- If we increase the number of cells in a circuit the bulb shines brighter
- When there are more light bulbs in a circuit the lighter is dimmer
- This is because each light bulb uses current the more they are the less the current will flow and the dimmer the light bulbs are.

2. MAGNETS AND MAGNETISM

Magnetism

Certain rocks contain iron and are called magnetite

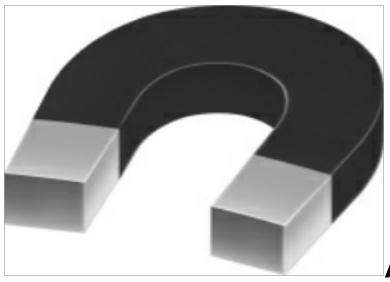
Magnets are made in different shapes

- Magnets are made in different shapes
- The most common are bar magnets



A bar magnet

- Horse shoe magnets



A horseshoe magnet

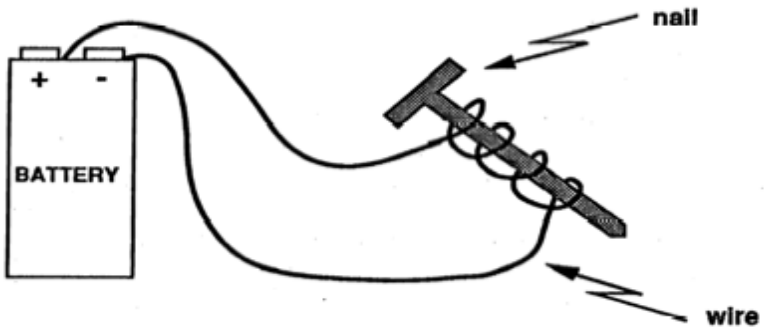
- They are called permanent magnets
- Because they do not lose their magnetism

Properties of magnets

- Some of the objects are attracted to the magnet (like those made out of iron and steel, or nickel)
- Any material that is attracted to a magnet is magnetic and a material that is not attracted by a magnet is called non-magnetic.

Electromagnetism

- Nearly 200 years ago a scientist discovered that when electricity current was flowing in a wire, there are magnetic forces around the wire. This was then used to build electromagnets.



3. DISEASES

What is a disease?

A condition that affects the proper function of the body or mind.

- A disease makes a person sick or unwell.
- May cause pain, tiredness or weakness.
- Causes rise in body temperature.
- May cause the sick person to die.

Causes of diseases

- Diseases are caused by small organisms called germs.

- There can only be seen under a microscope.
- Germs can go into our bodies in different ways, breathing, swallowing through the food we eat, getting into the blood stream when there is a cut in the skin or the person is bitten by a mosquito and ticks.

Communicable diseases

- A communicable disease is one where the germs can pass from the sick person to a healthy person. The healthy person can become sick with the disease.
- Tuberculosis is a communicable disease.
- HIV/AIDS is very dangerous communicable disease.
- Children disease like measles, chicken pox, mumps, diphtheria, whooping cough.

Non-communicable diseases

Malaria is a non-communicable disease

- The germs are carried from the sick person by a mosquito.
- Bilharzia is another non communicable disease.
- Cancer, diabetes and heart diseases are non- communicable diseases because they cannot be passed from one person to another.

4. SAFETY

Safety rules

- Our schools and homes are places where we feel safe but people can die.
- Accidents can be prevented if we learn about dangers which are around us.
- Safety rules that can stop accidents happening.

SAFETY RULES

- Make fire in a protected place where there is no wind.
- Do not let animals or children play next to a fire.
- Do not sit or stand close to a fire.
- Never let a fire burn in a room with all windows and doors closed the burning fire

will use all the oxygen and people will easily suffocate.

Putting out fire

A burning fire needs 3 things to keep burning;

- Heat to make the fuel hot enough to burn.
- Oxygen
- Fuel

a) Removing the heat

- If a fire is cooled down it will die out.
- This can be done by using water to cool it down
- If there are electric cables one should not spray water on the fire it will cause a short circuit that will make the fire worse.
- If its petrol burning one should not spray it with water but use sand.

b) Remove oxygen/air

- The easiest way to cut off fire is to cut air from the flames.
- One can wrap a blanket around the person whose clothes are burning.
- One can pour sand or dirt on to a fire.
- Where petrol or an electrical fire one cannot use water but use a fire extinguisher.
- If there is a fire in a room one should try keep windows and doors closed. This will cut off the flow of air to the fire

c) Remove any fuel that may burn

- If there is a gas turn it off.

ROAD SAFETY

Accidents are the second greatest cause of death in Botswana. Only AIDS kills more people.

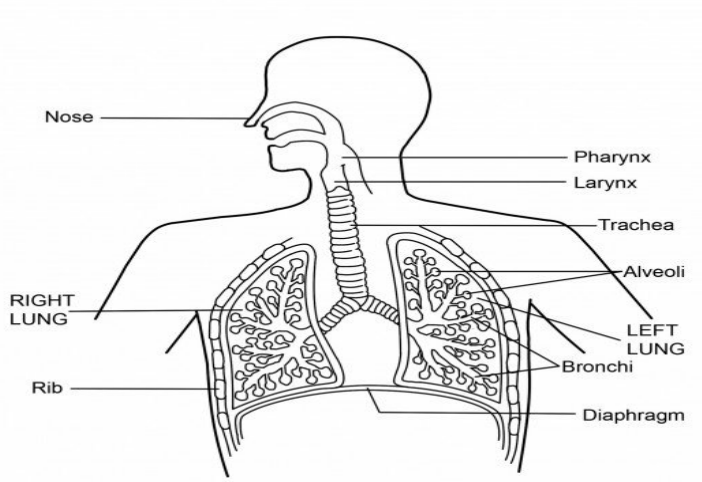
BODY SYSTEMS

Human body systems

- All living things are made up of building blocks and cells.
- There are so small that one can see them through a microscope.
- Cells in our bodies have different shapes.
- There are many different types of cells including body cells, liver cells, blood cells and bone cells.
- When the same cells are grouped together they are called tissue.
- The body has many tissues including muscle tissue, nerve tissue and blood tissue.
- When different tissues are put together they form an organ e.g the heart is made up of muscle tissue, nerve tissue and blood.
- When different organs work together it is called a system.
- The heart, arteries and veins together form a system called the circulatory system or blood system.

CELLS ↔ TISSUES ↔ ORGANS ↔ BODY SYSTEM

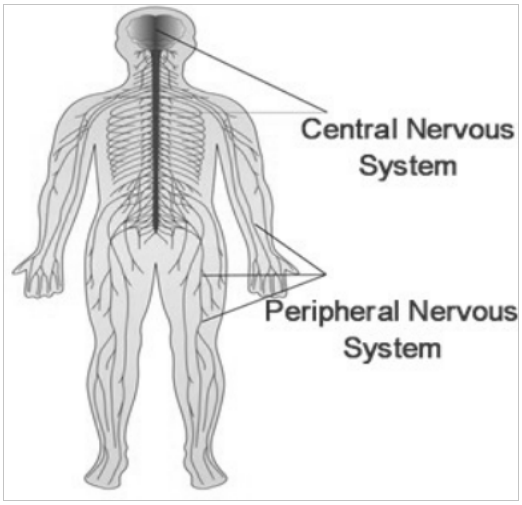
The respiratory system – these are organs with which we breathe, this brings oxygen into our bodies



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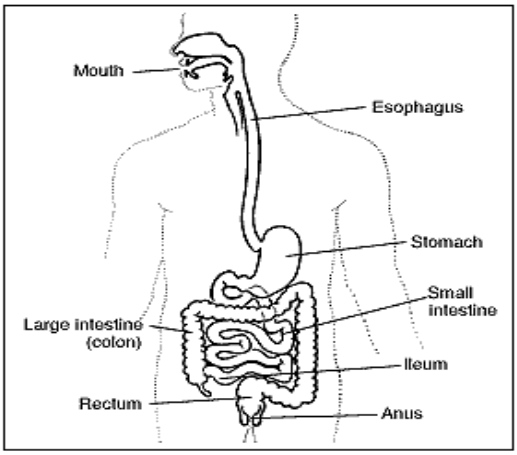
The skeletal system – these are the bones that keep the body firm and strong.

The nervous system – these are the brains and nerves they let us feel, see and hear things. The brain let us think.



The excretory system – these are the kidneys, bladder and other organs.

The digestive system – this system is just like a tube that starts from the mouth to the anus



The circulatory system – this is the system that circulates blood throughout the body. It is made up of the heart that pumps the blood through the tubes or blood vessels of the body.

The blood vessels which carry blood away from the heart are called the arteries. Those that carry blood to the heart are called veins.

The reproductive system – the main function of the reproductive system is to produce a baby. Man and female reproductive organs are different.

PHYSICAL DEVELOPMENT

PUBERTY

Puberty is the period when bodes of girls and boys change from that of a child to that of an adult and they can produce a child. We say they have become sexually mature the body is preparing for manhood and motherhood.

Signs of puberty in girls

- Skin becomes oily
- Breasts develop
- Body hair begins to grow in her armpits and on genitals
- The vagina becomes bigger

- The hips widen
- The ovaries start producing eggs
- The womb becomes bigger and monthly periods begin

Signs of puberty in boys

- The skin becomes oily
- Muscles develop more than in girls
- Broad shoulders develop
- Nipples may become painful
- Hair grows in armpits and on the chest
- The beard appears
- Pubic hair grows
- The voice breaks
- The penis grows
- The testes become painful and start producing sperms

Menstruation

- Menstruation happens when a woman releases blood and tissue from the womb. It happens once in 28 days.
- If the woman is not pregnant she will call it her periods.
- It usually lasts from 3 to 7 days.
- Menstruation starts during puberty. Menstruation stops during pregnancy and starts when the baby is born.
- After the age of approximately 45 years, menstruation usually stops permanently. This is called **menopause**